Lovenish

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.03
Lohiya Public Sr. Sec. School	Class XII (Rajasthan Board of Secondary Education)	2022	86.80%
Lohiya Public Sr. Sec. School	Class X (Rajasthan Board of Secondary Education)	2020	85.17%

TECHNICAL SKILLS

- Languages: C, C++, SQL, TypeScript, Python , Javascript

- Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

- Backend Concepts: REST APIs, Authentication, Authorization, RBAC, JWT, Database Transactions, ACID Properties,

Relational Database Design, Database Indexing, Query Optimization, Pagination, Input Validation, Error Handling,

Logging, API Security, Unit Testing, Integration Testing, Database Migrations

- Tools & Technologies: Node.js, Fastify, Prisma, PostgreSQL, MySQL, Docker, Git, GitHub, GitHub Actions, Linux CLI

PROJECTS

Backend Banking Management System -- C++, MySQL

- Built a transaction-safe banking management system supporting deposits, withdrawals, fund transfers, transaction history and administrative controls using database transactions, commit/rollback mechanisms and real-time balance updates.
- Designed relational schemas for users, accounts and transactions, applying ACID properties, constraints, prepared statements and transaction isolation concepts to maintain financial data consistency and prevent SQL injection.
- Implemented secure authentication and role-based access control with USER/ADMIN privileges, password protection, failed-login tracking, account freezing and OTP-based verification for sensitive fund transfers.
- Developed transaction monitoring with date, amount and user filtering, pagination, high-value transaction detection, administrative controls, audit logging and monthly financial analytics.

AI Document Q&A System | ***Python, LangChain, LLM, RAG, FAISS, Sentence Transformers***

- Developed a Retrieval-Augmented Generation (RAG) system to answer user queries using information retrieved from uploaded documents.
- Built the document pipeline for text extraction, chunking, embedding generation, and FAISS-based semantic retrieval.
- Engineered context-aware prompts that combine retrieved document context with user queries to generate grounded LLM responses.
- Optimized retrieval and response quality by experimenting with chunk size, top-k retrieval, and prompt configurations.

DevForge AI -- TypeScript, Node.js, Fastify, PostgreSQL

- Built a modular REST backend using TypeScript, Node.js and Fastify, organizing authentication, user and project management through controller, service and repository layers.
- Implemented JWT-based authentication and authorization with password hashing, protected routes, resource ownership checks and role-based access control for secure API operations.
- Designed PostgreSQL data models using Prisma ORM with relational mappings, constraints, migrations and indexed queries for reliable and efficient data access.
- Added schema-based request validation, centralized error handling, structured logging and environment-based configuration to enforce API contracts and improve backend reliability.

- Containerized the backend using Docker and configured GitHub Actions CI with automated testing, linting and build verification, while documenting APIs and development setup.

POSITIONS OF RESPONSIBILITY

Volunteer, Office of Career Services (OCS), IIT Hyderabad

- Assisted in coordinating and managing Online Assessments (OAs) and company recruitment processes, ensuring smooth execution of placement activities and timely resolution of operational issues.
- Supported company interviews and candidate coordination, helping maintain seamless communication between recruiters, students and the placement team during recruitment drives.

EXTRACURRICULAR

NSS Volunteer

- Contributed to social welfare initiatives, community outreach drives and public awareness campaigns, supporting organized on-ground activities and promoting community engagement.

Volunteer, Fresher Induction & Admission Support

- Assisted with student admission and orientation activities, coordinating documentation, campus tours and query resolution while helping ensure a smooth onboarding experience for incoming students.

RELEVANT COURSES

- **Computer Science & AI:** Introduction to Programming, Introduction to Modern AI, Machine Learning and its Applications, Modeling and Simulation

- Mathematics: Calculus I & II, Differential Equations, Elementary Linear Algebra, Introduction to Statistics, Introduction to Probability

- Mechanical Engineering: Engineering Mechanics, Mechanics of Solids, Fluid Mechanics, Heat and Mass Transfer,Control Systems, Manufacturing Science